**Winds:** Air flows from area of high pressure to area of low pressure. The warm air molecules over hot surface rise and create low pressure. So, cooler air with high pressure flows towards low pressure area. This causes wind flow.

**Chimneys:** Tall chimneys are kept in kitchen and industrial furnaces. As the hot gases and smoke are lighter, they rise up in the atmosphere.

### 7.2.3 Radiation

Radiation is a method of heat transfer that does not require particles to carry the heat energy. In this method, heat is transferred in the form of waves from hot objects in all direction. Radiation can occur even in vacuum whereas conduction and convection need matter to be present. Radiation consists of electromagnetic waves travelling at the speed of light. Thus, radiation is the flow of heat from one place to another by means of electromagnetic waves.

Transfer of heat energy from the sun reaches us in the form of radiation. Radiation is emitted by all bodies above 0 K. Some objects absorb radiation and some other objects reflect them.




While firing wood, we can observe all the three ways of heat transfer. Heat in one end of the wood will be transfered to other end due to conduction. The air near the wood will become warm and replace the air above. This is convection. Our hands will be warm because heat reaches us in the form of radiation.

**Radiation in daily life**

i. White or light coloured cloths are good reflectors of heat. They keep us cool during summer.
ii. Base of cooking utensils is blackened because black surface absorbs more heat from the surrounding.
iii. Surface of airplane is highly polished because it helps to reflect most of the heat radiation from the sun.

## 7.3 Concept of Temperature

Temperature is the degree of hotness or coolness of a body. Hotter the body, higher is its temperature.

### 7.3.1 Unit of Temperature

The SI unit of temperature is ***kelvin*** **(K)**. For day to day applications, ***Celsius*** (°C) is used. Temperature is measured with a thermometer.

### 7.3.2 Temperature scales

There are three scales of temperature.

i. Fahrenheit scale
ii. Celsius or Centigrade scale
iii. Kelvin or Absolute scale

**Fahrenheit scale**

In Fahrenheit scale, 32 °F and 212 °F are the freezing point and boiling point respectively. Interval has been divided into 180 parts.

**Celsius temperature scale**

In Celsius scale, also called centigrade scale, 0°C and 100 °C are the freezing point and boiling point respectively. Interval has been divided into 100 parts. The formula to convert a Celsius scale to Fahrenheit scale is:

$$F=\frac{9}{5}C+32$$

The formula for converting a Fahrenheit scale to Celsius scale is:

$$C=\frac{5}{9}(F-32)$$

**Kelvin scale (Absolute scale)**

Kelvin scale is known as the absolute scale. On the Kelvin scale 0 K represents absolute zero, the temperature at which the molecules of a substance have their lowest possible energy. The solid, liquid, gaseous



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